

EE 3202.304

ECE Fundamentals Lab II

Lab 3: BJT - AC

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10/13/2025

I. Experimental Results

Part 1 – Single Stage Amplifier

Amplifier Gain in Decibels (dB)

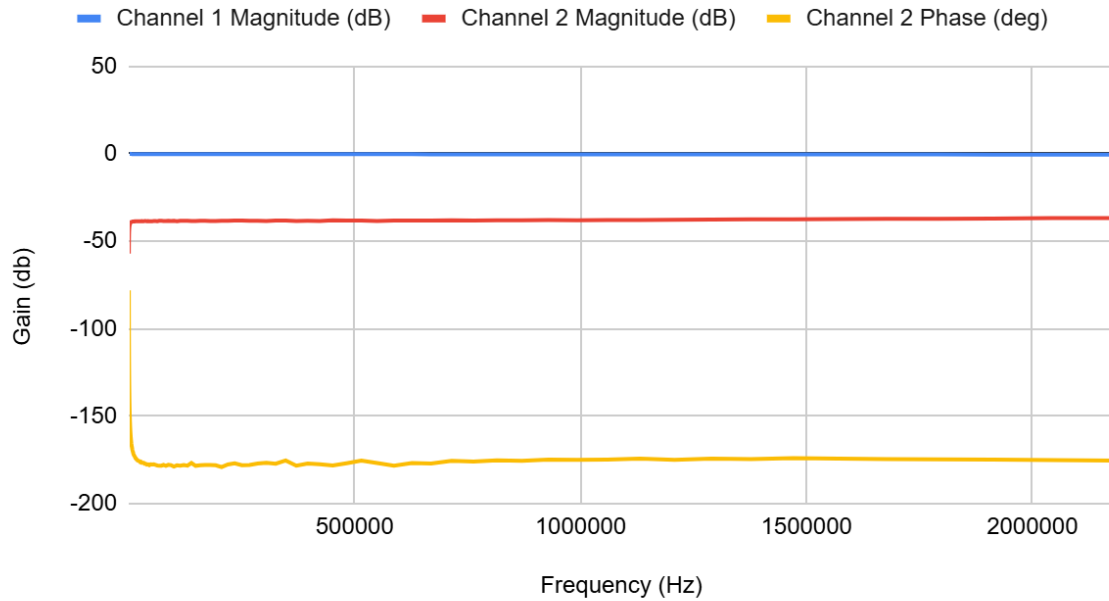


Figure 1.1: Amplifier Gain in DB vs Frequency

Amplifier Gain in Volts

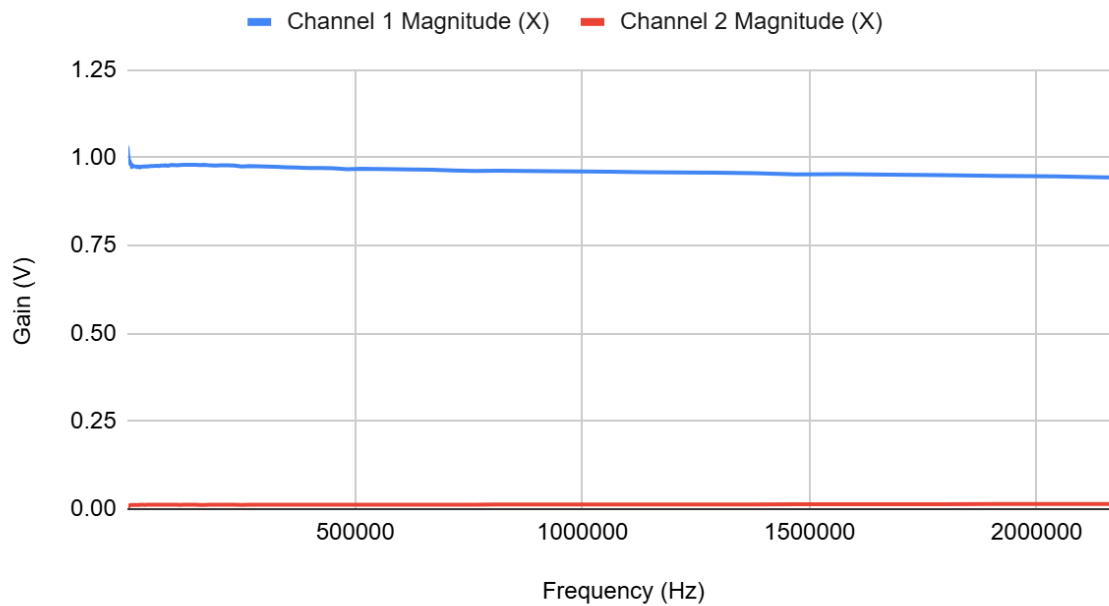


Figure 1.2: Amplifier Gain in Volts vs Frequency

Input and Output Voltage Response

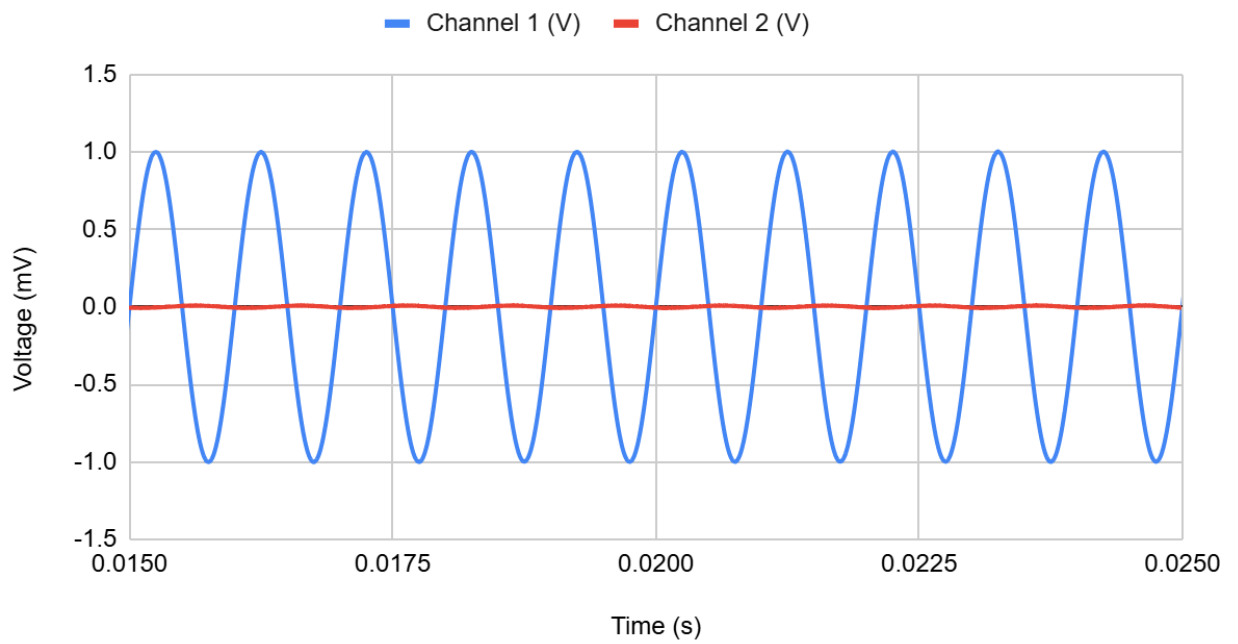


Figure 1.3: Amplifier Voltage Response

Part 2 – Differential Amplifier

Differential Amplifier of Opposite Polarities

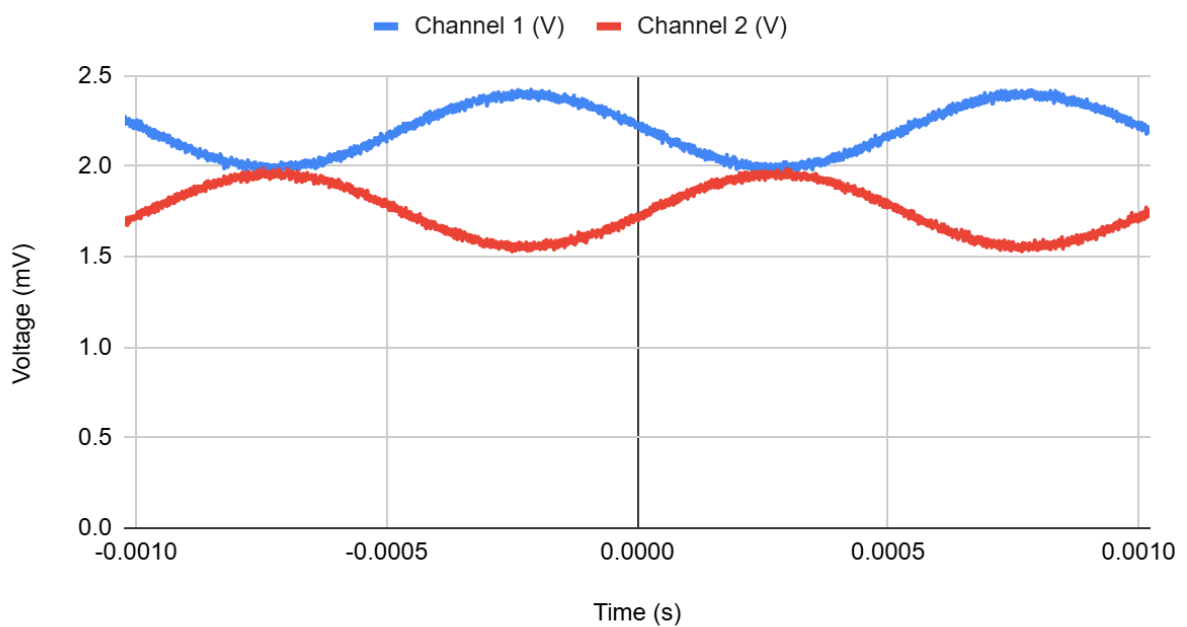


Figure 2.1: Differential Amplifier Voltage Response of Opposite Polarities

Math Channel = C1-C2/.02

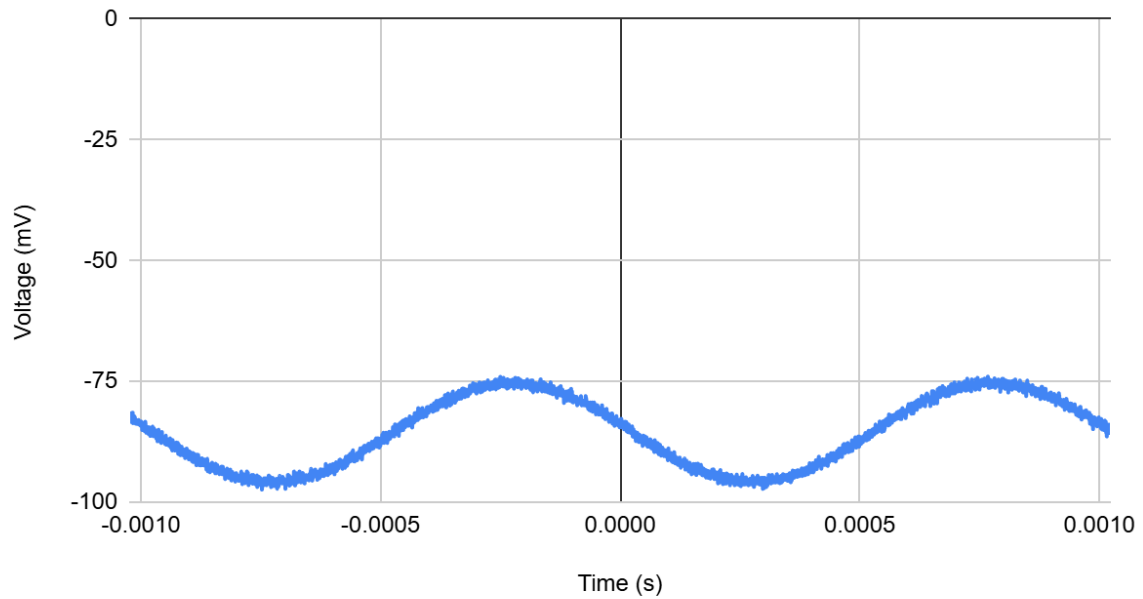


Figure 2.2: Differential Amplifier Math Channel

Channel	Measurement	Value
Channel 1	Peak 2 Peak	469.39 mV
Channel 2	Peak 2 Peak	462.11 mV
Math 1	Peak 2 Peak	23.552 V
Channel 1	Average	2.12 V
Channel 2	Average	1.80 V

Figure 2.3: Voltage Measurements Used for Impedance Calculations

II. Analysis and Thoughts

Part 1

The single-stage amplifier built in part 1 shows voltage amplification using a BJT in the active region. In this configuration, the transistor operates as a voltage-controlled current source, where small variations in the base-emitter voltage produce large changes in collector current, resulting in a voltage gain across the collector resistor. We can also see that there is a 180-degree phase difference between the input and outputs.

From the pre-lab design, the expected midband voltage gain was around -8 . Experimentally, the measured gain was close to this theoretical value in the mid-frequency range around 5 kHz and 20 kHz. At low frequencies, the gain began to drop due to the coupling capacitors limiting low-frequency response acting like a high-pass filter. Similarly, at higher frequencies above 30 kHz and 40 kHz, the parasitic capacitances in the transistor and wiring caused a slight gain drop, forming the upper cutoff frequency of the amplifier.

The frequency gain curve, in **Figure 1.1 and 1.2**, showed a relatively flat midband region where the amplifier maintained constant gain, confirming that the transistor was properly biased in the active region. When the input amplitude was too large, clipping and distortion were observed in the output waveform, indicating that the transistor was entering saturation or cutoff, where it no longer behaves linearly. By reducing the input amplitude, a clean sinusoidal output was restored.

The minor discrepancies between simulated and measured gains likely stemmed from component tolerances, breadboard parasitic, and probe loading effects from the oscilloscope. Despite these, the circuit's overall behavior matched the theoretical predictions, successfully demonstrating how a properly biased BJT can act as a small-signal voltage amplifier with predictable midband gain and frequency limits.

Part 2

In the second part of the lab is building a BJT differential amplifier. This circuit amplifies the difference between two input voltages (V_{in+} and V_{in-}) while rejecting any common-mode signals, such as noise or interference that appear equally on both inputs. Conceptually, the differential amplifier relies on a shared emitter resistor or current source that enforces signal coupling between the two transistors. When one input increases while the other decreases by the same amount, current shifts between the two transistors, producing a large voltage swing at the output. In contrast, when both inputs move together the total current remains nearly constant, leading to minimal output change.

In the oscilloscope measurements in **Figure 2.1 and 2.2**, the two input signals (C1 and C2) were set to equal amplitudes but opposite polarities. This produced two output waveforms that were 180° out of phase with each other, showing how each transistor responded oppositely to the differential input. To visualize the actual differential output, the math

function $(C1 - C2)/0.02$ was used. This operation subtracts one channel from the other and divides by the input amplitude (0.02 V peak, or 20 mV), giving the voltage gain of the differential amplifier as seen with our math channel. This calculation represents how much the circuit amplified the difference between the two input signals.

To determine the input impedances, we can apply a series resistor with one of the differential inputs to get the input current. Once we have our input current, we can easily find our input impedance using Ohms law. As seen in **Figure 2.3** with a table of our values.

Experimentally, the measured differential gain was slightly lower than theoretical value due to transistor mismatches and resistor tolerances, while the input impedance differed somewhat from simulation because of real transistor base currents and parasitic effects. At higher frequencies, a small gain reduction was seen from internal transistor capacitances and shared emitter impedance, both of which limit bandwidth. Despite these differences, the results confirmed that the circuit amplified the voltage difference as expected demonstrating the operation of a differential amplifier.

III. References

None